

Progesterone and enhanced intelligence

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THE THALIDOMIDE disaster has made everyone aware of the possible injurious effect on the fetus caused by certain drugs administered during pregnancy. It is therefore reassuring to learn of a drug, administered during pregnancy, which appears to give a positively beneficial effect by improving the intellectual potential of the unborn child.

In the 1960s, work by Cheftez in America, simultaneously with my work in London, suggested that progesterone might have such a positive effect on the fetus. Cheftez examined mothers' obstetric histories in a sample of mentally defective children and concluded that all too frequently there was a suggestion of hormonal imbalance during the pregnancy and, in particular, evidence of progesterone deficiency.

Meanwhile I had observed in my general practice that children of mothers who had received progesterone for the relief of pregnancy symptoms, or as prophylaxis against threatened or habitual abortion or pre-eclamptic toxæmia, tended to do well at school. At one time five such children were top of their respective forms at the local school.

Above average

Before any such hypothesis can be accepted, it must be subjected to rigorous testing. First there was a pilot study in which the local headteachers were asked to assess progesterone children and matched-control children of the same age and sex, and 55 per cent of progesterone children were considered to be above average compared with 41 per cent of the matched-control children.

At this time, controlled trials into the use of progesterone injections in the prophylaxis of pre-eclampsia were proceeding at Chase Farm Hospital, Enfield, in which expectant mothers who volunteered were allocated at 16 to 28 weeks of pregnancy to either a progesterone treatment or a symptomatic treatment group. The children from these trials (which *did* show significantly that progesterone had a place in the prevention of pre-eclampsia) were followed up at their first birthday examination at the local clinic by the medical officers or health visitors.

This survey showed that although there was no difference in talking or teething, more children in the progesterone group were standing and walking than among the control children. Furthermore, the results showed that the effect of progesterone was dose dependant and that those progesterone children whose mothers had received high dosage made better progress than those who only received low dosage.

With such encouraging results a follow-up was organised of children born in the City of London Maternity Hospital between 1955 and 1957, at a time when progesterone had been used for the relief of pregnancy symptoms (headaches, nausea and vomiting, depression, lethargy, irritability, and fainting) and in the prevention of pre-eclampsia.

The children were then 9 to 10 years old and each progesterone child was matched with a control, the next child born in the labour ward register whose mother had a normal pregnancy, and the next mother who developed pre-eclamptic toxæmia. With the parents' permission the headteachers were approached and asked for an assessment of the children in English, mathematics, verbal reasoning, craftwork, and physical education.

Again, more progesterone children were assessed as "above average" than control children, in all academic subjects: ie, English, mathematics, and verbal reasoning. The children of toxæmic mothers had the poorest assessments, which is in agreement with previous studies. Once again it was noticed that the higher the dose and the earlier the progesterone was administered, the better the academic achievements.

However, it was possible that the headteachers were biased towards the extrovert progesterone children, so the children were followed up again when they were 18-20 years old and an assessment made of their educational attainments. It was found that 59 per cent of progesterone children stayed on at school until they were 18, compared with 46 per cent of control children and 33 per cent toxæmic children.

More progesterone children obtained O- and A- levels than control children, while the toxæmic children obtained very few. Finally, of 34 progesterone

children, 11 (32 per cent) got a place at university compared with only 6 per cent of the control children, and this compares with 6 per cent of children getting university places in the Inner London Education Authority and 6 per cent in the Borough of Haringey in which most of the children lived.

Again, in this survey, it was noticed that the best results at O- and A- levels were among those children who received a high dose of progesterone, in whom progesterone was administered before the 16th week, or in whom the administration lasted for more than eight weeks. It was also interesting that at O- and A- levels the progesterone children preferred science subjects to the arts, and looking back it was noticed that at 9 to 10 years old more of the progesterone children were assessed as above average in mathematics compared with English.

Two psychologists, J. U. Zussman and P. P. Zussman, from Stamford University, California, were sent over to examine the progesterone children and they also confirmed that prenatal progesterone administration enhanced performance in mathematical problems and that this was related to the dose, time of administration, and duration of progesterone. The synthetic progestogens are known to cause masculinisation of the female fetus, but the Zussmans found that in contrast there was no evidence of masculinisation among these progesterone children.

Critical period

Exactly why progesterone should have this beneficial effect on brain growth is unknown, but it is recognised that a critical period exists during pregnancy and early postnatal life when malnutrition, even for short periods, may produce irreversible damage. In rats, malnutrition during fetal life interferes with cell division, producing a deficit in the number of cells in all organs, which persists even after re-feeding.

In the brain of the rat different regions show a selective sensitivity to malnutrition which appears to depend on the cellular events occurring in that region at that time. Conversely, "over-nutrition" produces an increased brain weight associated with an increased number of cells. It is possible that in the human fetus nutrition with progesterone increases the number of brain cells, which persists into adult life.

Much work still remains to be done to understand more fully the action of progesterone on the human fetus, and for the optimum dose, time, and duration of treatment to be found. Indeed, there may yet come a time when in addition to supplying all expectant mothers with iron and vitamin supplements, consideration will be given to the routine administration of progesterone.